Name:_____ Period:_____

The Fastest Thing Ever Built

Learning Intention: Today we will be learning how objects move in their orbits around the Sun so we can predict how fast an object travels and how long its year will be.

Success Criteria:
- I can describe an orbit as an ellipse with the Sun at one focus.
- I can use evidence to explain why an object moves fastest when it is closest to the Sun.
- I can use distance data to predict the length of a planet's year.

Part 1: The Puzzle

On December 24, 2024, NASA's **Parker Solar Probe** flew just 3.8 million miles above the surface of the Sun at **430,000 miles per hour**. That is faster than any human-made object has ever moved.

It did not get that fast by firing a giant engine. It got that fast by **falling**.

Here is the strange part: Earth is already moving sideways around the Sun at about **67,000 mph**, and every spacecraft we launch starts out moving with it.

1. Which do you think is harder: sending a spacecraft **into the Sun** or sending it **out of the solar system**? Circle one, then explain.

Into the Sun Out of the solar system

Why? _____

_____

2. Parker's orbit is a stretched-out oval. Where in its orbit do you predict it moves fastest? Circle one.

Closest to the Sun Farthest from the Sun Same speed the whole way

Part 2: Draw an Orbit

a. Lay the graph paper on the cardboard. Push two pins into the middle of the paper, **____ cm apart**.

b. Put the string loop around both pins. Pull it tight with your pencil and trace all the way around. This is **Orbit 1**.

c. Move the pins to **____ cm apart**, centered in the same spot, and trace again. This is **Orbit 2**.

d. For Orbit 1, label one pin hole **SUN**. Write **CLOSEST** at the point on the orbit nearest the Sun and **FARTHEST** at the point farthest away.

3. Which orbit is more stretched out (more **eccentric**)? _____

4. What happens to the shape of an orbit as the two pins (the **foci**) move closer together?

_____

5. Is the Sun at the center of Orbit 1? Explain._____

_____

Part 3: Equal Slices, Unequal Speeds

Kepler discovered that a line from the Sun to a planet sweeps out **equal areas in equal amounts of time**. Let's see what that means for speed.

Use the Equal-Area Diagram. An object takes the same amount of time to move from one numbered dot to the next.

6. Where are the dots far apart, and where are they close together?_____

_____

7. Where does the object travel the farthest in the same amount of time? Near the Sun or far from the Sun?

_____

8. Shade the slice from the Sun to dot 1 and dot 2 in one color. Shade the slice from the Sun to dot 7 and dot 8 in another color. Count the squares in each slice. Count a square if at least half of it is shaded.

Slice	Number of squares
Sun, dot 1, dot 2	
Sun, dot 7, dot 8	

a. What do you notice about the two areas?_____

b. Go back to your prediction in Question 2. Were you right? Use your slices as evidence.

_____

_____

Part 4: Find the Pattern

Planet	Mean distance from Sun (million km)	Time to orbit the Sun (one "year")
Mercury	57.9	88.0 days
Venus	108.2	224.7 days
Earth	149.6	365.26 days (1 year)
Mars	228.0	1.9 years
Jupiter	778.5	11.9 years
Saturn	1432.0	29.4 years

Planet	Mean distance from Sun (million km)	Time to orbit the Sun (one "year")
Uranus	2867.0	84.0 years
Neptune	4515.0	164.8 years

Data: ESRT Solar System Objects Data Table

9. What pattern do you see between distance from the Sun and the length of a year?_____

_____

10. Jupiter is about **5 times** farther from the Sun than Earth. Is its year 5 times longer? What does that tell you?

_____

11. Predict: The dwarf planet **Ceres** is 414 million km from the Sun. About how long is its year? Explain how you decided.

My prediction: _____ Because: _____

_____

Now check your ESSRT. Actual answer: _____

12. Parker Solar Probe now circles the Sun about once every **3 months**, close to Mercury's year. Use the pattern to explain why its year is so short.

_____

Part 5: The Goldilocks Zone

The **habitable zone** is the range of distances from a star where a planet could have **liquid water** on its surface. Too close, and water boils away. Too far, and it freezes. Earth sits inside the Sun's habitable zone. Many scientists' estimates put Mars near the outer edge, yet Mars's water is frozen or locked underground.

13. Why does a planet's distance from its star affect whether it can have liquid water?_____

_____

14. Mars is near the outer edge of the habitable zone, but it has no liquid water on its surface. What else besides distance might matter? (Think about atmospheric conditions)

_____

Part 6: Happy Birthday, Neptune

To find your age on another planet, divide your age in Earth years by that planet's year (in Earth years). Mercury's year is **0.24** Earth years.

Planet	My age there
Mercury (0.24 years)	
Mars (1.9 years)	
Neptune (164.8 years)	

15. No person alive today has lived through one full Neptune year. Use the pattern from Part 4 to explain why.

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